End-to-End Azure Data Engineering Project Using Medallion Architecture

Project Summary

This project demonstrates the implementation of a modern cloud-based data engineering solution on Microsoft Azure. The objective was to design and develop an automated, scalable, and maintainable data pipeline capable of ingesting raw CSV data, transforming it through multiple processing layers, and generating analytics-ready datasets for reporting and downstream consumption.

The solution follows the Medallion Architecture (Bronze, Silver, and Gold layers) and leverages Azure-native services for orchestration, storage, processing, and data management.

Business Objective

Organizations often receive large volumes of raw data from multiple sources that require cleansing, transformation, and standardization before they can be used for analytics and reporting.

The primary objective of this project was to:

- Automate data ingestion from source files.
- Establish a structured data lake architecture.
- Improve data quality through transformation and validation.
- Implement incremental data processing to reduce execution time and costs.
- Deliver analytics-ready datasets in an optimized format.

Solution Architecture

Source Layer

- Source data is received in CSV format.
- Files are placed in the designated landing zone for processing.

Ingestion Layer

- Azure Data Factory is used to orchestrate and automate the ingestion process.
- Source files are copied into the Bronze layer of Azure Data Lake Storage Gen2.

Processing Layer

- Azure Databricks notebooks perform data transformation and validation.

- PySpark is used for scalable data processing.

Storage Layer

- Azure Data Lake Storage Gen2 is used to store data across Bronze, Silver, and Gold layers.
- Parquet format is used to optimize storage and query performance.

Technology Stack

Service	Purpose
Azure Data Factory	Pipeline orchestration and scheduling
Azure Data Lake Storage Gen2	Centralized data storage
Azure Databricks	Data transformation and processing
PySpark	Distributed data processing
Azure SQL Database	Metadata validation and lookup operations
GitHub	Source control and version management

Data Processing Workflow

Bronze Layer – Raw Data Storage

The Bronze layer stores raw ingested data exactly as received from the source systems.

Key Activities:

- Data ingestion using Azure Data Factory.
- Raw file archival.
- Schema preservation.
- Historical data retention.

Output:

- Raw CSV data stored in ADLS Gen2 Bronze container.

Silver Layer – Data Cleansing and Standardization

The Silver layer focuses on improving data quality and preparing datasets for business transformations.

Key Activities:

- Data type conversions.
- Null value handling.
- Duplicate record removal.
- Data validation checks.
- Schema standardization.

Output:

- Clean and validated datasets stored in ADLS Gen2 Silver container.

Gold Layer – Business Ready Data

The Gold layer contains transformed datasets optimized for analytics and reporting.

Key Activities:

- Business rule implementation.
- Dimension table creation.
- Data enrichment.
- Aggregation and transformation logic.
- Incremental processing.

Output:

- Analytics-ready Parquet datasets stored in ADLS Gen2 Gold container.

Incremental Data Loading Strategy

To improve efficiency and reduce processing overhead, incremental loading techniques were implemented.

Key Benefits:

- Processes only newly added or modified records.
- Reduces compute resource consumption.
- Improves pipeline execution performance.
- Supports scalable data growth.

Data Storage Optimization

The final datasets are stored in Parquet format.

Advantages:

- Columnar storage architecture.
- Improved compression ratio.
- Faster analytical queries.
- Reduced storage costs.
- Optimized performance with Spark workloads.

Monitoring and Error Handling

The solution includes operational monitoring and error tracking mechanisms.

Implemented Features:

- Pipeline monitoring through Azure Data Factory.
- Databricks job execution tracking.
- Activity-level failure logging.
- Retry mechanisms for transient failures.
- Execution status monitoring.

Version Control and CI/CD Readiness

GitHub integration was implemented to support:

- Version control.
- Source code management.
- Collaborative development.
- Change tracking.
- Deployment readiness.

Key Achievements

- Developed an end-to-end Azure Data Engineering solution.
- Implemented Medallion Architecture using Bronze, Silver, and Gold layers.
- Automated data ingestion and transformation workflows.
- Built scalable PySpark-based processing pipelines.
- Implemented incremental loading for optimized performance.
- Delivered analytics-ready datasets in Parquet format.

Conclusion

This project successfully demonstrates the design and implementation of a production-style Azure Data Engineering solution. By combining Azure Data Factory, Azure Data Lake Storage Gen2, Azure Databricks, and PySpark, the solution provides a scalable and efficient framework for ingesting, transforming, and managing enterprise data while following modern data engineering best practices.